Roll No:.....

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Amritapuri

MID - TERM EXAMINATION OCT 2025

Third Semester

Int MSc Physics & Mathematics
28th October 2025 – AN

24CSA210 – Programming for Scientific Computing

Duration: 2 Hrs Maximum : 50 Marks

INSTRUCTIONS TO CANDIDATES

1. This examination paper contains TEN (10) questions and comprises FOUR (4) printed pages.
2. Answer all questions. The maximum marks for first five questions is THREE (3) each. The next three questions are FIVE (5) marks each. The last two question are assigned TEN (10) marks each.
3. Answer the last question beginning on a FRESH page of the answer book.
4. Candidates may use calculators for back calculation.

Question 1. (3 marks) (CO1, L1)

(i) Why is python called an interpreter language?

(ii) Give two main differences between a compiler and interpreter language with examples of each type?

Question 2. (3 marks)(CO2-L3,CO3-L1)

(i) Write the syntax for list slicing. How is list slicing useful in extracting data from lists?

(ii) Define three different errors encountered in a python program.

Question 3. (3 marks)(CO2-L2,L4)

(i) Explain for loop and While loop with syntax.

(ii) How can you reduce the use of loops to process arrays in python?

Question 4. (3 marks)(CO2-L1,L2)

(i) What is Numpy package used for? How is Numpy package faster than the array package?

(ii) What are the three important components of a Numpy array?

Question 5. (3 marks)(CO2-L1,L2)

(i) What are tuples? What is the difference between a tuple and a list?

(ii) What package performs advanced scientific calculations? What is the advantage of using this package over numpy?

Question 6. **(5 marks)(CO4-L1,CO5-L3)**

(i) How do you handle errors in python? Explain with syntax and examples.

(ii) What is probability and why do you need the concept of probability? How do you simulate probability values in python?

Question 7. **(5 marks)(CO2-L3)**

What is Matplotlib library and what is it used for? Rewrite the following equation as a python function:

$$y = 18x^3 + 5x$$

Use the Matplotlib library to plot this equation for x values in the range 1,2,$\cdots$,5?

Question 8. **(5 marks)(CO4-L1,CO1-L1)**

(i) What are the modes of opening a file in python? Give five file types (extensions) that can be modified using file handling.

(ii) What is the syntax for the 'with' function and how is it useful in file handling?

Question 9. **(10 marks)(CO1-L5)**

How can coding actually simplify the human effort? What are the reforms that have come after the discovery of programming and how do you think scientific computing can actually reform our lives? Write about this elaborately.

Question 10. **(10 marks)(CO2-L3)**

Write the full program for the following problem and add the code to plot the results.

A ball is dropped vertically from rest from the top of a tall building. Ignoring air resistance, the motion of the ball can be described by the equation of motion under constant acceleration:

$$s = ut + \frac{1}{2}gt^2$$

where:

s = **displacement (in meters)**
u = **initial velocity (m/s)**
g = **acceleration due to gravity** (9.8 **m/s**^{2})
t = **time (in seconds)**

Initial Conditions

- **The ball starts from rest, so** $u = 0$.
- **Acceleration due to gravity,** $g = 9.8$ **m/s**^{2}.
- **The motion is observed for** $t = 0$ **to** $t = 5$ **seconds.**

Plot a graph of displacement (s) on the y-axis versus time (t) on the x-axis.

Enjoy!!!